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Pharmacology of Antihypertensives, Part 1:

Renin Angiotensin System Antagonists (RAS Inhibitors)

- Angiotensin Converting Enzyme Inhibitors
- Angiotensin II AT_1 Receptor Blockers
- Direct Renin Inhibitors

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completing the preparation materials, students should be able to:

1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the ACE inhibitors, AT1 receptor blockers, and direct renin inhibitors to their effects on vascular tone, renal function, and cardiovascular structure by giving examples of therapeutic applications.
3. Compare and contrast the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug’s pharmacologic effects when presented with a clinical case vignette.

Preparation Materials (links are in the CPG and on the next slide)

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Required

- ScholarRx Bricks | Practice Questions and Clinical Vignettes

Highly relevant optional materials:

- Video lecture | Dr. Goldstein's Word handout | Guided Reading Questions (GRQs)
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

Scholar Rx Brick (Required)

Antihypertensive drugs https://exchange.scholarrx.com/brick/antihypertensive-drugs

Renin-Angiotensin-Aldosterone System Inhibitors https://exchange.scholarrx.com/brick/renin-angiotensin-aldosterone-system-inhibitors

Helpful bricks on blood pressure physiology (Recommended but not required):

Autonomic regulation of the cardiovascular system https://exchange.scholarrx.com/brick/autonomic-regulation-of-the-cardiovascular-system

Hypertension: Foundations and Frameworks https://exchange.scholarrx.com/brick/hypertension-foundations-and-frameworks

Primary (Essential) Hypertension https://exchange.scholarrx.com/brick/primary-essential-hypertension

Baroreceptor regulation of blood pressure https://exchange.scholarrx.com/brick/baroreceptor-regulation-of-blood-pressure

Humoral regulation of blood pressure https://exchange.scholarrx.com/brick/humoral-regulation-of-blood-pressure

Textbooks/Exam review books you might find helpful:

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024

Chapter 11: Antihypertensive Agents

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281748109

Chapter 17: Vasoactive Peptides > Angiotensin

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281749431

LWW Health Libraries Premium Basic Sciences Lippincott Illustrated Reviews: Pharmacology, 8e, 2022; Chapter 8: Antihypertensives

https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325981&bookid=3222

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 11: Drugs Used in Hypertension

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285590661

Questions help learning. Questions help to master a topic.

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1. **Guided Reading Questions** are intended to help you identify what you NEED to know.
   - Other information in the reading provides context to help you understand concepts within the big picture.
2. **Practice Questions** are for your own assessment of your learning – for applying what you have learned in the context of clinical case vignettes.
   - Practicing through case vignettes will help you bridge pharmacology science and pharmacotherapeutics (clinical concepts).
3. **Write down your own questions** as you study. This practice helps you identify where you are strong and where you weak so you can focus your efforts.

Brown, Roediger, and McDaniel. *Make It Stick The Science of Successful Learning*. Belknap Press of Harvard University Press, 2014

Tips for effective LEARNING (not rote memorizing).

- Identify and define key ideas/concepts.
- Rephrase MAIN ideas in your OWN WORDS.
- Convert MAIN points to questions.
- Relate the ideas to what you already know.

Techniques:

- Spacing your practice – reviewing material and questions after a period of time improves learning by giving your mind time to make connections.
- Mixing multiple subjects (interleaving) while you are studying improves learning and problem solving skils by forcing the brain to continually retrieve knowledge.
- Use / invent memory devices (mnemonics) to help you remember.

Brown, Roediger, and McDaniel. *Make It Stick The Science of Successful Learning*. Belknap Press of Harvard University Press, 2014

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What you need to learn about:

Prerequisites: To understand the mechanisms and effects of therapeutic interventions with blood pressure lowering drugs (the pharmacology of antihypertensive agents), students should have an understanding of normal cardiovascular functions: hemodynamics, systolic and diastolic blood pressure, baroreceptor and humoral regulation of blood pressure, autonomic nervous system response, and the renin-angiotensin-aldosterone system.

- The recommended blood pressure goals
- The antihypertensive agents preferred for initial therapy: calcium channel blockers, angiotensin converting enzyme inhibitors (ACEIs), angiotensin AT-1 receptor blockers (ARBs), and thiazide and thiazide-like diuretics – and why they are preferred initial drugs for the management of primary hypertension and their relative efficacy
- What antihypertensive agents are considered “second-line” and why they are used as adjuncts to the initial drugs

What you need to know

- The mechanism of action of each of the classes of antihypertensive agents
- How each of the “starred” drugs reflects the class characteristics and what properties make them different from the rest of the drugs in the class
- The reflex (compensatory) baroreceptor and humoral responses to the initial and chronic therapy with specific classes and doses of antihypertensive agents
- The patient factors and the potential drug-disease interactions that should be considered when selecting antihypertensive therapy for the individual patient
- The clinically relevant pharmacokinetic properties:
  - By what route will the patient take the medication?
  - How often will the patient have to take it?
  - Can it be given intravenously for the treatment of hypertensive emergencies?
  - What are the potential drug-drug interactions and their mechanisms?
- The additional non-hypertension uses of each class of drugs
- What to tell the patient when counseling them about their medications

Background: Regulation of Blood Pressure

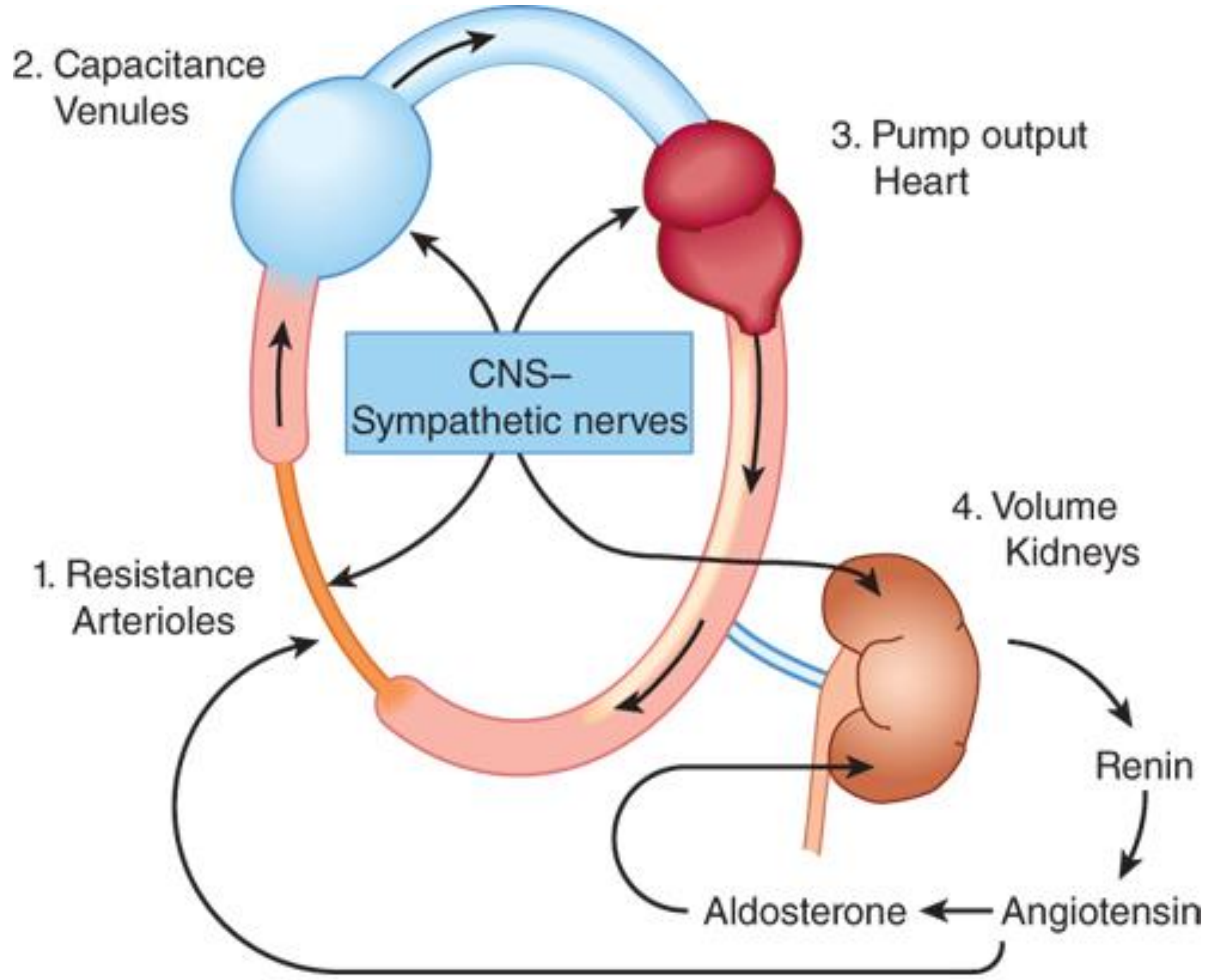
All antihypertensive drugs produce effects by interfering with normal mechanisms of BP regulation:

- Resistance arterioles
- Capacitance venules,
- Heart (cardiac output),
- Kidneys (volume)

➢ Antihypertensive therapy can induce compensatory responses to a drop in blood pressure:

    Increased sympathetic outflow

    Increased renin secretion



Anatomic sites of blood pressure control.

Antihypertensive agents work by reducing peripheral resistance, cardiac output, and/or volume.

Categories:

1. RAS inhibitors reduce PVR and potentially blood volume.
2. Direct vasodilators reduce PVR.
3. Sympatholytics inhibit cardiac function and increase pooling in capacitance vessels.
4. Diuretics deplete body sodium and reduce blood volume (and possibly other mechanisms).

RAS: renin-angiotensin system

PVR: peripheral vascular resistance

Mc Graw Hill

Citation: Chapter 11 Antihypertensive Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021, Figure 11-1. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595338 Accessed: June 13, 2023

Slide adapted by LG.

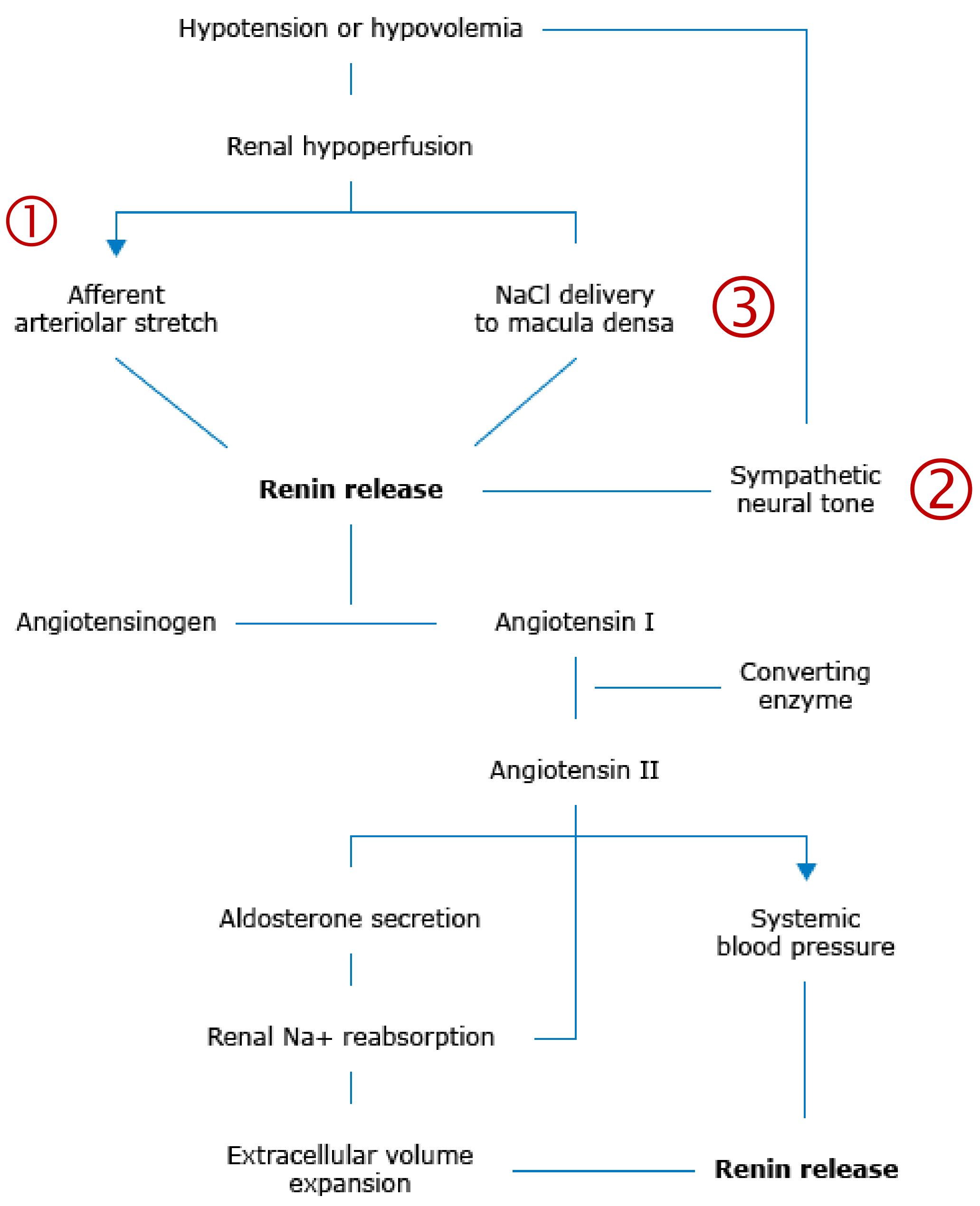
Determinants of arterial pressure:
Cardiac output and peripheral resistance
BP = CO x SVR

- Mean arterial pressure
  - Cardiac output
    - Stroke volume
    - Heart rate
  - Peripheral resistance
    - Vascular structure
    - Vascular function

small arteries and arterioles

BP: blood pressure
CO: cardiac output
SVR: systemic vascular resistance

Regulation of renin release



The classic renin-angiotensin-aldosterone system and the maintenance of sodium and volume balance.

UpToDate®

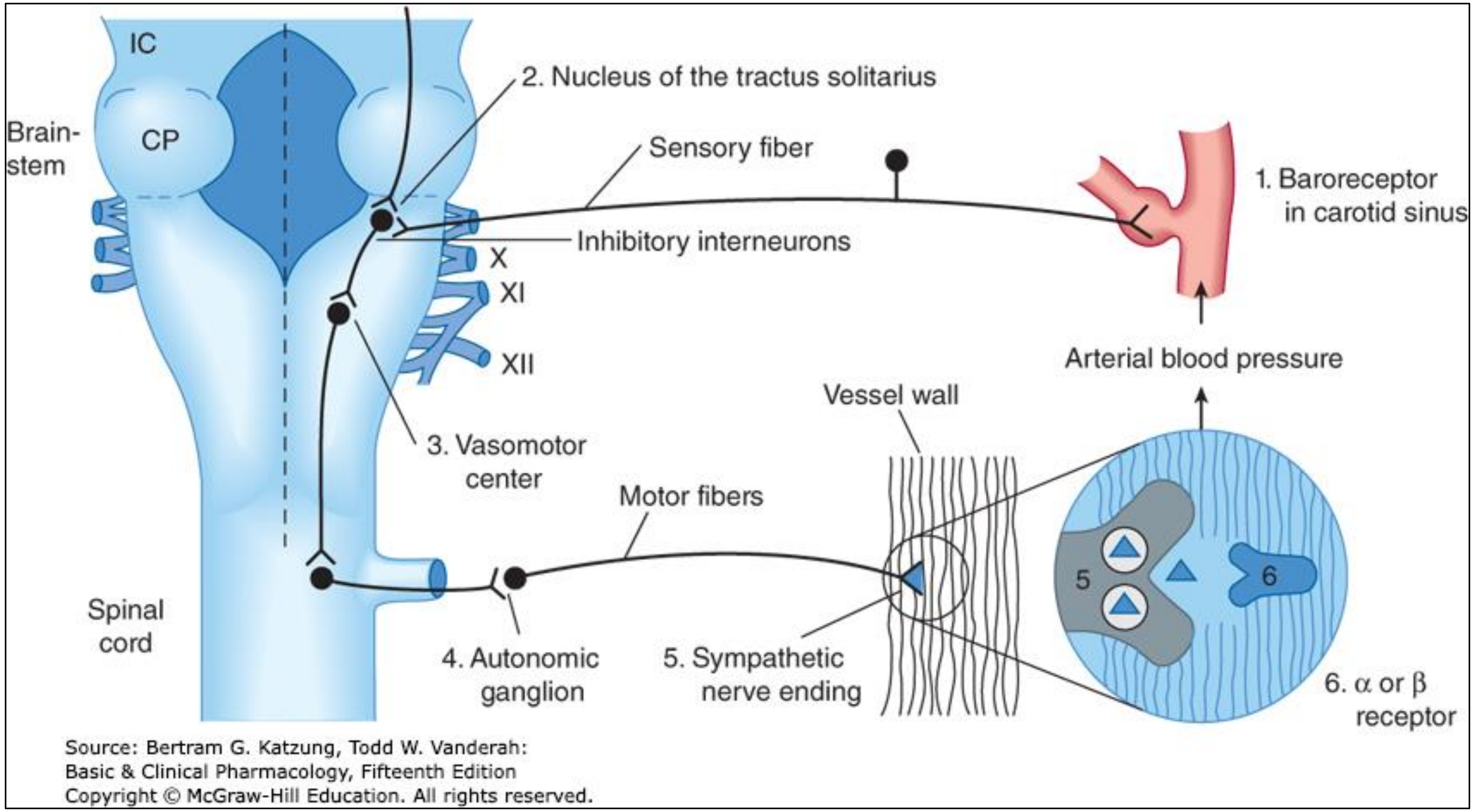
Changes in extracellular fluid volume are sensed by:

- (1) Baroreceptors in the wall of afferent arteriole sense relaxation of vascular smooth muscle, which enhances renin secretion.
- (2) Cardiac and arterial baroreceptors and circulating catecholamines stimulate β_1ARs, which enhance renin secretion.
- (3) Reduction in NaCl delivery to the cells of the macula densa enhances renin secretion.

Slide adapted by LG

Graphic 65116 Version 3.0

Baroreflexes provide rapid, moment-to-moment adjustments in blood pressure.



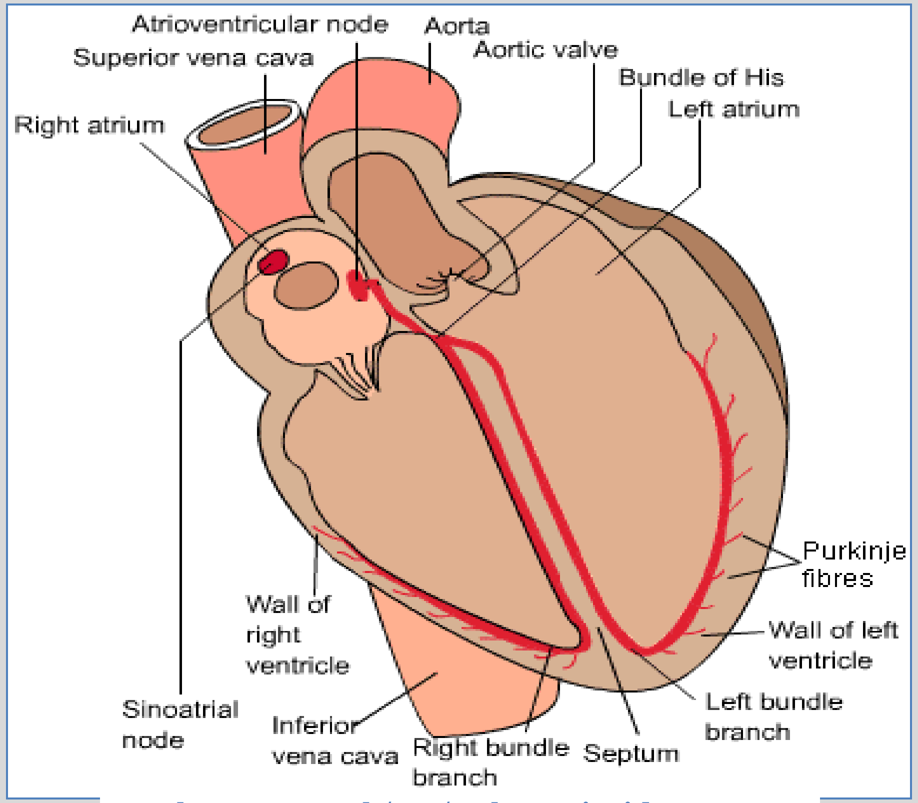
- Central sympathetic neurons from the vasomotor area of the medulla are tonically active.
- Carotid baroreceptors stimulated by vessel wall stretch inhibits central sympathetic discharge.
- Baroreceptors that sense reduction in arterial pressure (reduced wall stretch) disinhibit sympathetic discharge.
- Reflex increase in sympathetic outflow increases PVR and CO.

The kidney provides long-term blood pressure control by controlling blood volume.

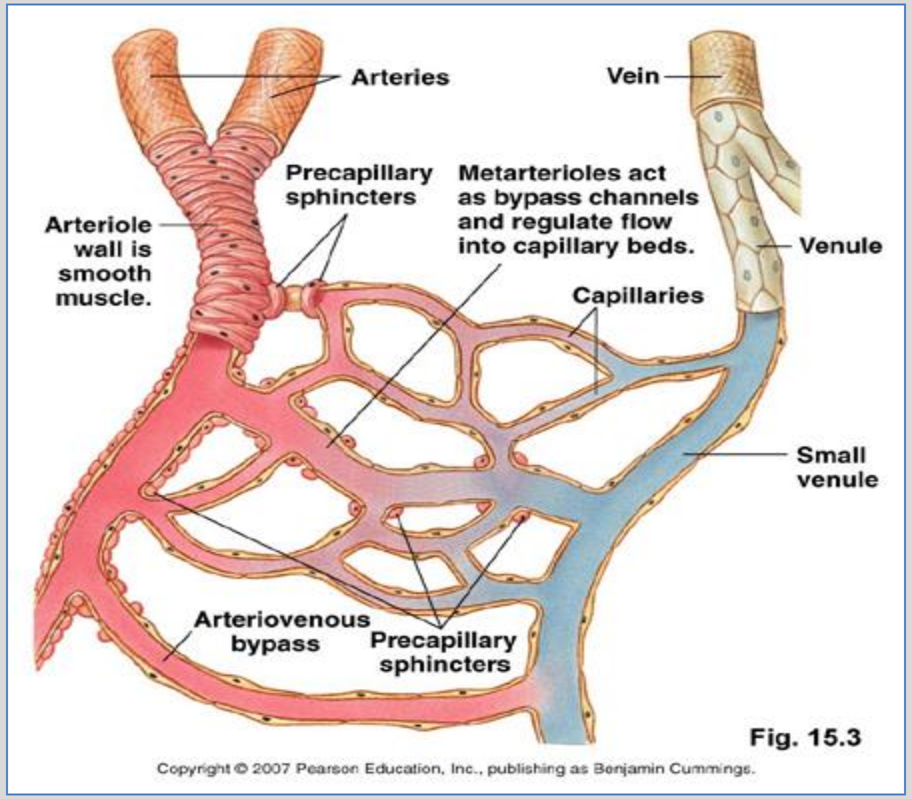
Baroreceptor reflex arc. CP, cerebellar peduncle; IC, inferior colliculus.

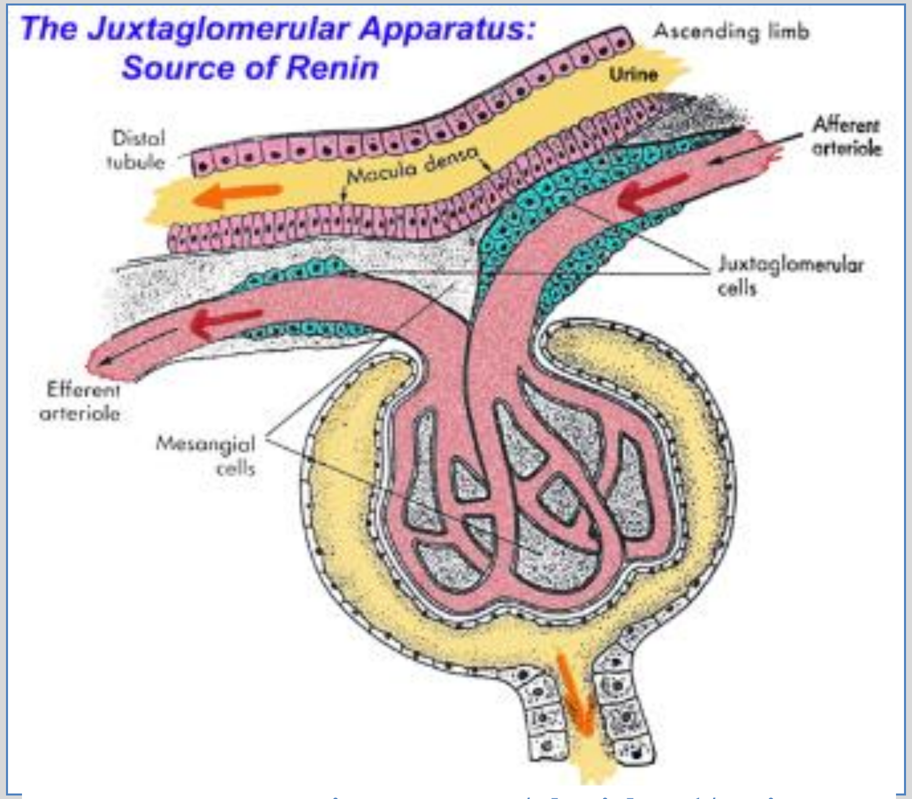
Citation: Chapter 11 Antihypertensive Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021, Figure 11-2. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595338 Accessed: June 13, 2023

Question: How does each of the following systems contribute to blood pressure regulation?



hsc.uwe.ac.uk/rcp/cs-heart-inside.aspx





www.marcoantonioramos.com/physiology1/reninse...

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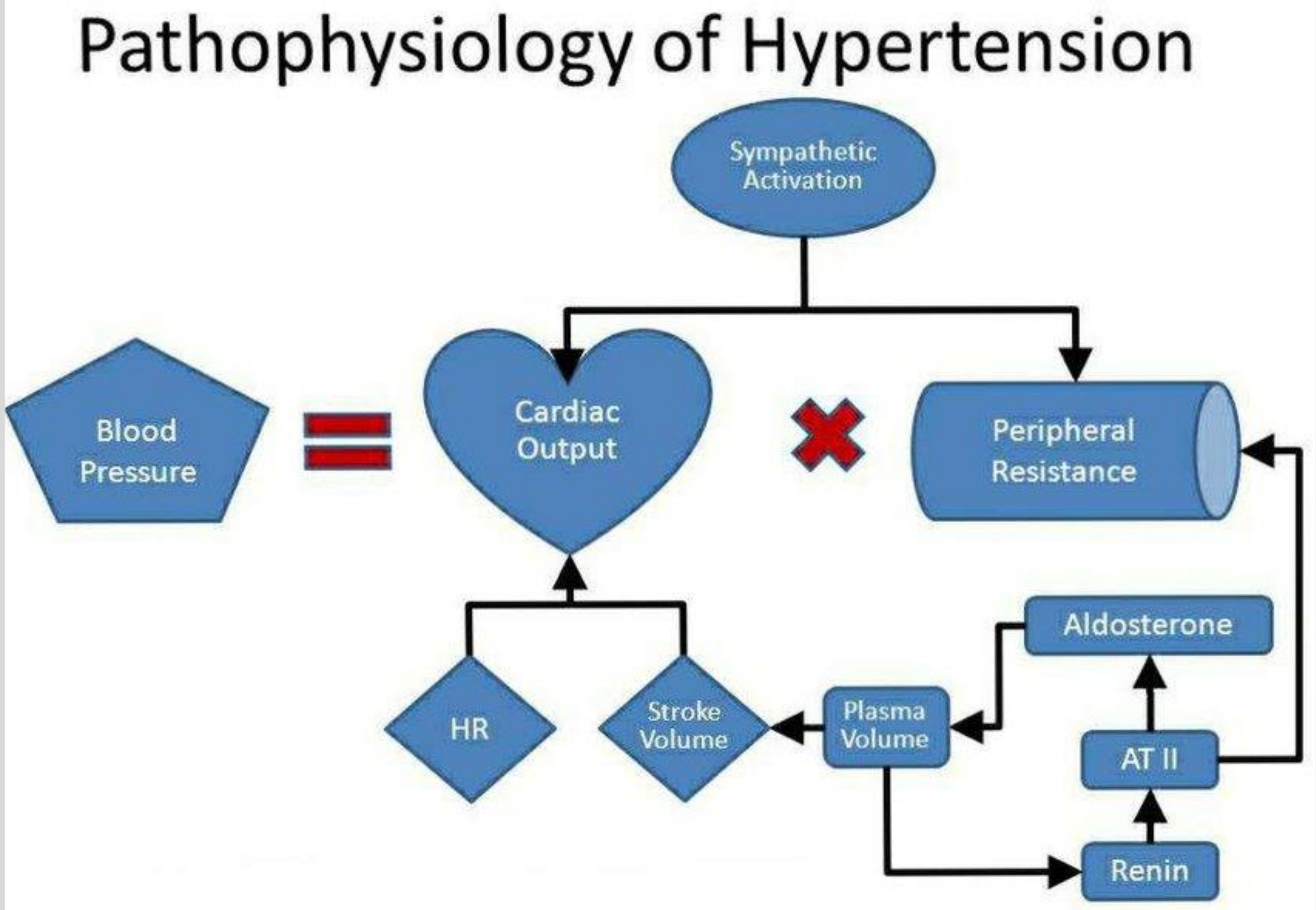

https://www.researchgate.net/figure/Fig-Pathophysiology-of-Hypertension_fig2_340579663

Treatment of Hypertension: Goals

Lifestyle modifications

Drug therapy

Individualize Antihypertensive Therapy

2025 ACC/AHA High Blood Pressure Guidelines

Blood pressure categories:

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- Normal: Less than 120/80 mm Hg;
- Elevated: Systolic between 120-129 *and* diastolic <80;
- Stage 1: Systolic between 130-139 *or* diastolic between 80-89;
- Stage 2: Systolic at least 140 *or* diastolic at least 90 mm Hg;
- Hypertensive crisis: Systolic over 180 and/or diastolic over 120, with patients needing prompt changes in medication if there are no other indications of problems, or immediate hospitalization if there are signs of organ damage.

2025 ACC/AHA High Blood Pressure Guidelines Screening

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CVD risk factors in adults with hypertension:

- smoking,
- excessive weight,
- diabetes,
- dyslipidemia,
- low fitness, unhealthy diet, psychosocial stress, sleep apnea

HTN evaluation

- target organ damage
- established CV or renal disease
- CV risk factors
- lifestyle contributors to HTN
- drugs:

sympathomimetics, NSAIDs, drugs containing Na^+, erythropoietin, angiogenesis inhibitors

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Question: Integrating osteopathic principles and practice and OMT in patient care, including pharmacotherapy.

- A 42-year-old person presents with new onset hypertension. What areas might you examine for viscerosomatic reflexes to help you decide if this patient's issue is primary or secondary?

Rational treatment is based upon an understanding of the basic principles of body unity, self-regulation, and the interrelationship of structure and function.

NYITCOM OMM Faculty

Pharmacology of Hypertension: Antihypertensive Agents

Classes of Antihypertensive Agents

		Diuretics (see lecture on diuretics pharmacology)
Antihypertensive Agents Video Lectures	Part 1	Angiotensin converting enzyme inhibitors (ACEIs) Angiotensin II AT1 receptor antagonists (ARBs) Renin inhibitors
	Part 2	Calcium channel blockers Beta blockers
	Part 3	Alpha blockers Sympathoplegics (centrally acting) Sympathetic nerve terminal blockers Vasodilators (other)

Antihypertensive Drugs Actions and Classes

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Preferred initial therapy	Second-line therapy options
– Angiotensin converting enzyme inhibitors – Angiotensin II AT_1 receptor blockers – Calcium channel blockers (CCB), long-acting – Thiazide-type / thiazide-like diuretics	– Beta blockers – Alpha blockers – Alpha-2 agonists – Vasodilators – Direct renin inhibitor

Initial Antihypertensive Therapy General Recommendations

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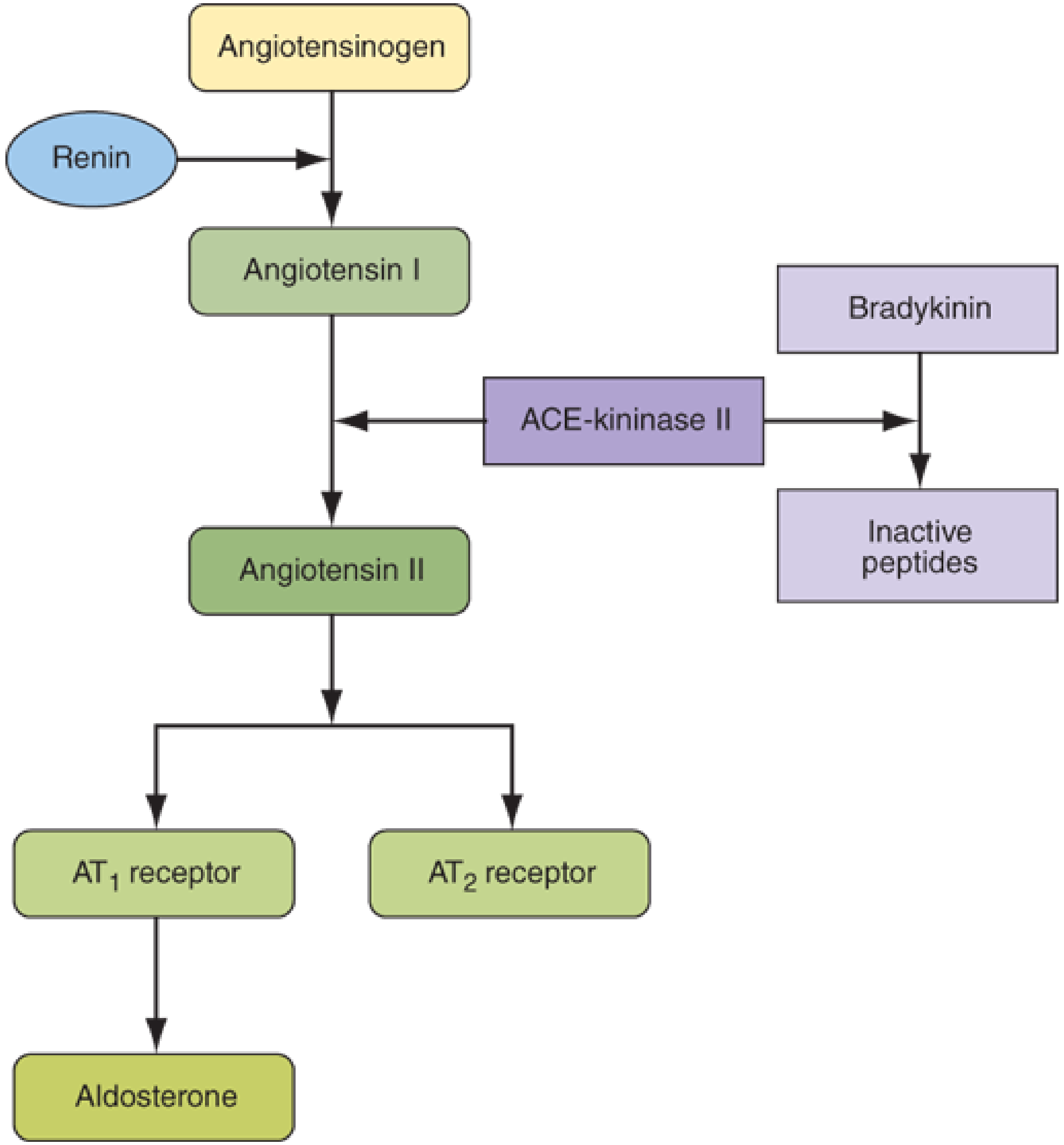
The attained blood pressure is the major outcome determinant of reduction in CV risk – not the choice of drug. Therapy should be patient-specific.

Monotherapy: Stage 1 HTN 130-139/80-89 mmHg	Combination therapy: BP >20/10mm Hg above goal or Stage 2 HTN or SBP ≥150, DBP≥90 mmHg (Experts differ in their approaches.)
• ACEI or ARB or CCB • Thiazide / thiazide-like diuretic for patients with edema, osteoporosis, calcium nephrolithiasis	Recommendation: • ACEI or ARB + CCB (ACCOMPLISH trial) • ACEI or ARB + thiazide/thiazide-like • CCB + thiazide/thiazide-like diuretic

For patients who require initial combination therapy with two drugs, they should be preferred agents with complementary mechanisms of action, low incidence of side effects, and long-acting, unless the individual patient has an indication for a drug from a different class.

ACEI: ACE inhibitor; ARB: angiotensin receptor blocker; DHP CCB; dihydropyridine calcium channel blocker

RAS Inhibitors:
Angiotensin Converting Enzyme Inhibitors (ACEIs)
and
Angiotensin II AT1 Receptor Blockers (ARBs)

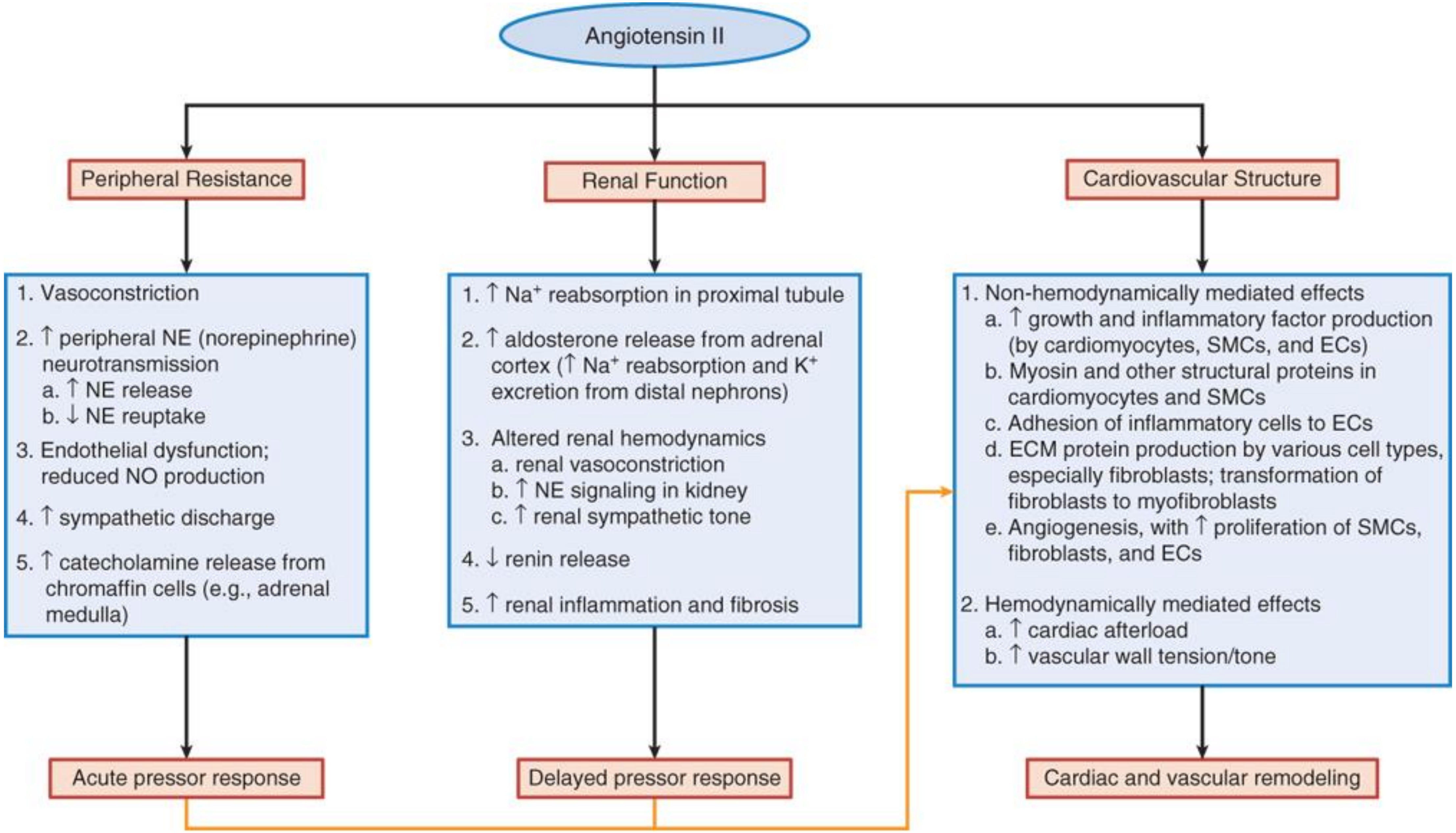


Renin–Angiotensin–Aldosterone System

1. Renin: hydrolyzes angiotensinogen to ANG I
2. ANG I $\xrightarrow{ACE}$ ANG II
3. ANG II has potent vasoconstrictor properties via the AT1 receptors.
4. Aldosterone causes Na^+ and water reabsorption

Renin-angiotensin-aldosterone axis. ACE, angiotensin-converting enzyme.

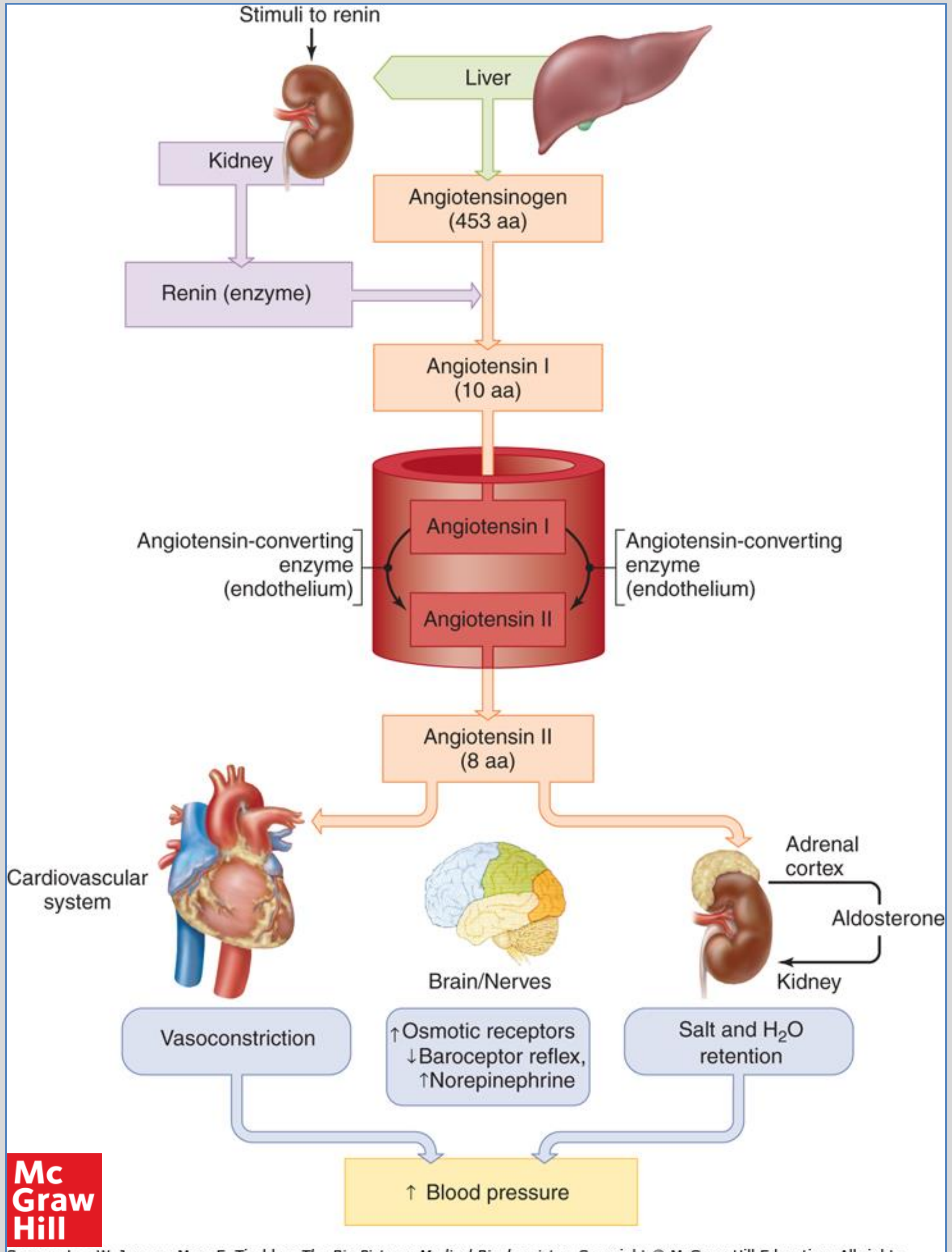
Citation: Chapter 277 Hypertension, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=265454202&bookid=3095&Resultclick=2 Accessed: June 09, 2023



Major physiological effects of AngII. AngII exerts physiological and pathological effects on both the renal and cardiovascular systems, providing a rationale for the therapeutic benefits of RAS inhibition. EC, endothelial cell; SMC, smooth muscle cell.

Citation: Chapter 30 Renin and Angiotensin, Brunton LL, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e;* 2023, figure 30-5. Available at: https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=269721276&gbosContainerID=0&gbosid=0&groupID=0&sectionId=265828708&multimediaId=undefined Accessed: June 09, 2023

Effects of renin, angiotensin II and aldosterone on organs controlling volume status and blood pressure



The Big Picture: Medical Biochemistry 2018: Chapter 18. The Urinary System; Figure 18-4.

Angiotensin AT1 and AT2 Receptors

- **AT1 receptors**

$G_{q/11}$PCRs $\rightarrow$ PLC activation

$\downarrow$

$\uparrow$ cytosolic Ca^{2+}, stimulates PKC

(1) vasoconstriction

(2) hypertrophy

(3) fibrosis, inflammation

(4) nephropathy

- **AT2 receptors**

$G_{i\alpha 2/3}$PCRs $\rightarrow$ $\downarrow$cAMP

and activate various phosphatases

(1) vasodilation

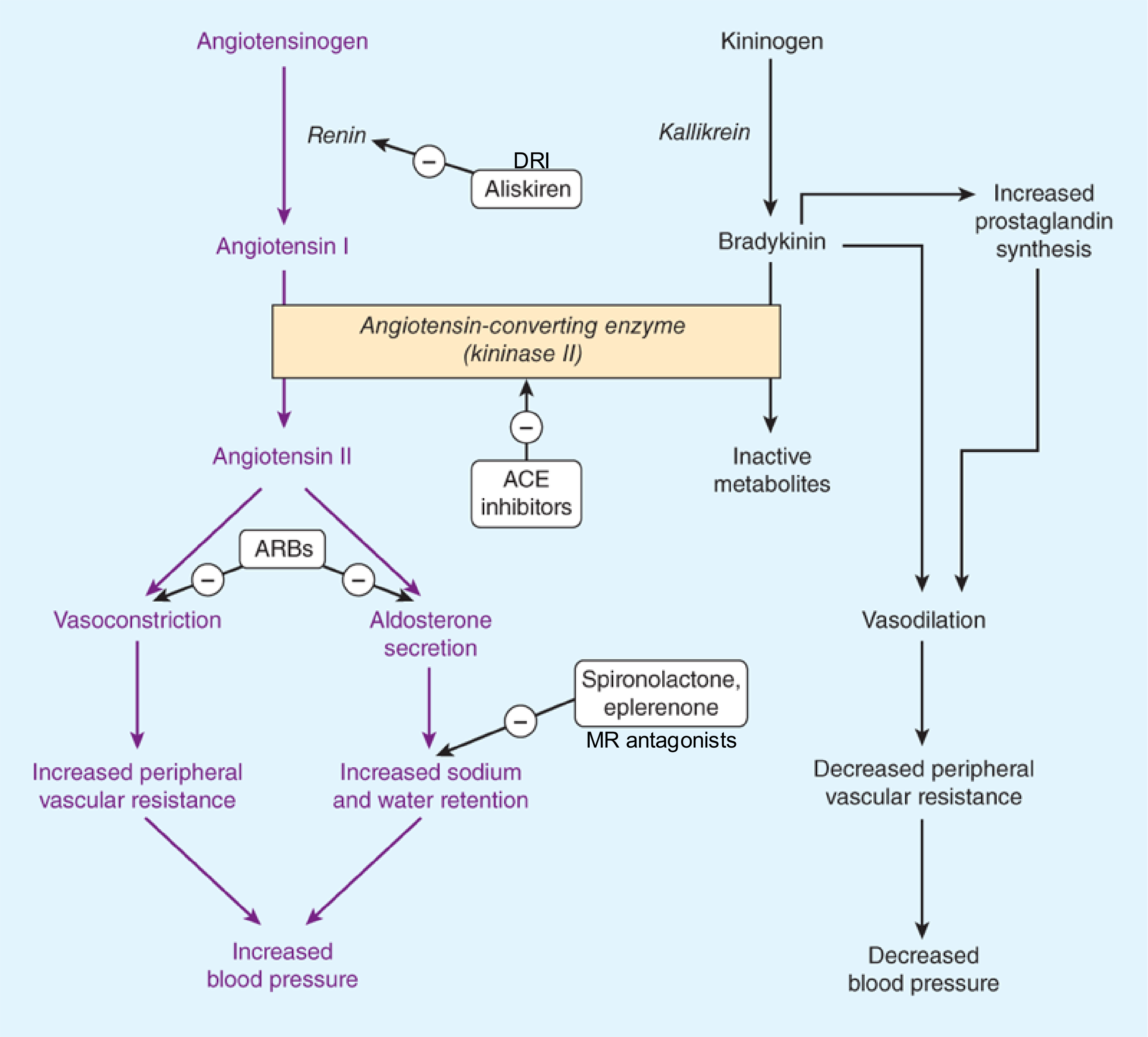
(2) natriuresis

(3) antagonize growth effects

AT1Rs include other GPCR types and are subject to downregulation.

AT2R include non-GPCR pathways.

Other receptor types mediate ANG1-7 effects.



Sites of action of drugs that interfere with the RAAS

ACE, angiotensin-converting enzyme

ARBs, angiotensin receptor blockers

DRI, direct renin inhibitor

MR, mineralocorticoid receptor

RAAS: renin-angiotensin-aldosterone system

Citation: **Chapter 11 Antihypertensive Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021, Figure 11-5**.
Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595338 Accessed: August 03, 2021

Inhibitors of Angiotensin II

Most ACE inhibitors are small ester-containing peptide prodrugs hydrolyzed by hepatic esterases to active molecule. Active metabolites are much more potent ACEIs than the parent compounds.

Most ACEIs are prodrugs		ARBs
Parent drug	Active molecule	
* Captopril	not a prodrug	Losartan *
* Enalapril	Enalaprilat	Valsartan
Fosinopril	Fosinoprilat	Azilsartan
Benazepril	Benazeprilat	Candesartan
Lisinopril	not a prodrug	Eprosartan
Moexipril	Moexiprilat	Irbesartan
Perindopril	Perindoprilat	Olmesartan
Quinapril	Quinaprilat	Telmisartan
Ramipril	Ramiprilat	
Trandolapril	Trandolaprilat	

ARBs are small molecule nonpeptide drugs.

Pharmacokinetics: General Class Properties

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	ACE inhibitors	ARBs
A	Oral prodrugs, generally moderate absorption Enalaprilat is IV Active compounds have poor absorption.	Oral, Good absorption
D	Moderate to extensive protein binding	Highly bound to plasma proteins
M	Hepatic esterases to active metabolite (except captopril and lisinopril - not prodrugs) Some are further metabolized.	Minimal hepatic metabolism Some are minor substrates of CYP2C9 Some have active metabolite
F	Bioavailability varies	Bioavailability range: 10-80% Irbesartan 60-80%; the rest have <50%
E	Urine and/or feces Some ACEIs require renal dosing adjustment	Feces and urine Some ARBs require renal dosing adjustment
$t_{1/2}$	Serum half-life varies	Range ~6 hours-12 h (adults) Longer in elderly
	Peak concentrations vary widely, 1-10 hours Captopril 2 to 3 doses per day. All others may be given once daily for HTN mgmt	Onset: 1-2 hours 1x daily dosing for HTN mgmt

Actions

ACE inhibitors	ARBs
Competitive inhibitors of angiotensin converting enzyme – Inhibits conversion of ANG I to ANG II – Reduces ANG II effects at both AT1 and AT2 receptors – Inhibits bradykinin and substance P degradation (vasodilating peptide / neuropeptide)	Competitive antagonists of angiotensin II AT1 receptor inhibitors – High-affinity binding – Permit activation of AT2 receptors – Stimulate renin release, which increases circulating ANG II levels, but ARBs block ANG II effects at AT1 receptors

Effects of both ACEIs and ARBs:

- ↓ arterial and venous pressure and peripheral vascular resistance; reduces preload, afterload
- heart rate and cardiac output change very little

  may cause reset of baroreceptor reflex to lower arterial pressure
- ↑ natriuresis by increasing renal blood flow and reducing aldosterone and ADH secretion
- ↓ resistance in efferent glomerular arteriole → ↓ glomerular capillary pressure and ↓ proteinuria
- prevent remodeling of smooth muscle and cardiac myocytes

ACEIs and ARBs provide similar benefits in patients with hypertension, heart failure, and chronic kidney disease.

Efficacy is enhanced by concomitant diuretic therapy and reduced dietary sodium intake (as with other RAS inhibitors).

ACEIs / ARBs: Use in Hypertension

ACEI or ARB are included in Guidelines as a preferred initial therapy option

- Alone or in combination with other antihypertensive agent(s)
- Start with low dose
- Potential for a large initial drop in blood pressure (hypotension), especially in patients who may have an active RAS
  - (e.g. diuretic-induced volume contraction or congestive heart failure)

Efficacy: Lower blood pressure to some extent in most patients with hypertension

- Full effect usually occurs after several weeks

Left ventricular hypertrophy regression is reported with ACEIs, ARBs.

2025 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines

ACEIs / ARBs: Additional Therapeutic Uses

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ACEIs may have benefits in addition to lowering the blood pressure in:

Post-myocardial infarction	Beneficial effects on mortality after acute MI ACEIs induce sustained BP lowering, regression of ventricular hypertrophy, and prevent ventricular remodeling after myocardial infarction; other mechanisms may contribute.
Heart failure with reduced ejection fraction	Improve symptoms, reduce hospitalizations, and prolong survival • With beta blocker (loop diuretic for fluid control in overt HF) ACEI, ARB, or neprilysin inhibitor-ARB (secubitril-valsartan)
Proteinuric chronic kidney disease, diabetic and nondiabetic	Slow progression of disease Reduce glomerular pressure by reducing BP and cause dilation of efferent arterioles Reduce protein excretion

ACEIs / ARBs: Adverse effects

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Effect	Mechanism
Hypotension	Usually occurs with first few doses Most often observed in a RAAS-dependent state, such as volume-depleted patients
Reduced GFR	Relaxed efferent arteriole Usually modest; may be severe in patients whose intrarenal perfusion pressure is already reduced.
Hyperkalemia	ACEIs and ARBs reduce aldosterone secretion, which impairs K+ excretion in the urine. (ANGII, also plasma K+ elevation, stimulate aldosterone secretion.

ACEIs / ARBs: Adverse effects

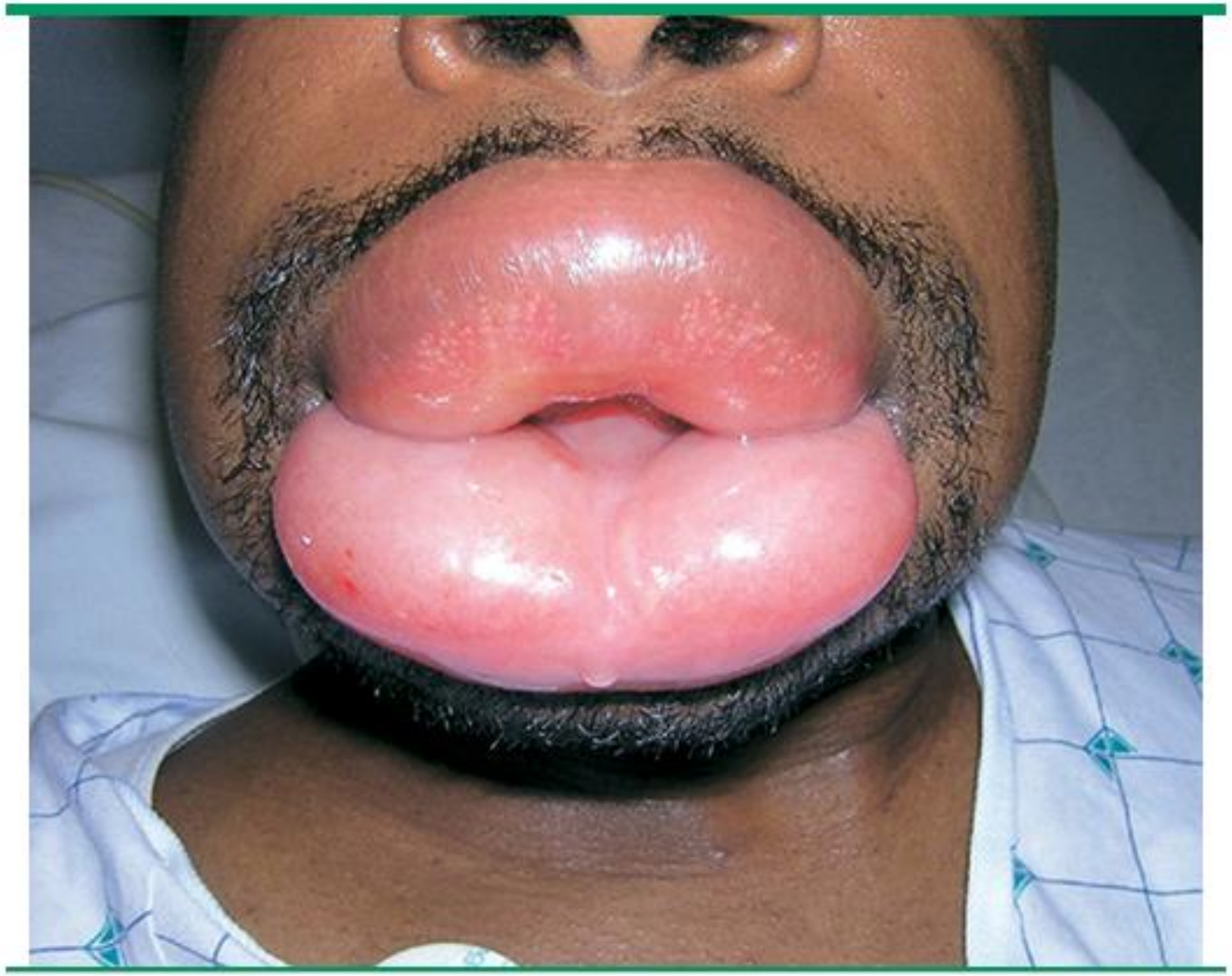
Effect	Mechanism
Dry cough	ACEIs: 5% to 20% of patients accumulation in lungs of bradykinin, substance P or prostaglandins Low incidence with ARB therapy.
Angioedema	ACEIs: Rare (0.1% to 0.7%), potentially fatal Bradykinin and substance P seem to be involved (inflammatory vasoactive peptides → vasodilation) ARBs: Incidence rates are not higher than with non-RAS antihypertensive drugs.
Enteropathy, sprue-like	Olmesartan characterized by severe chronic diarrhea and weight loss

Angioedema lip

Angioedema is a diffuse, nonpitting, tense swelling of the dermis and subcutaneous tissue. It develops over minutes to hours, and resolves over subsequent hours or days. Angioedema typically does not itch, unless it is associated with urticaria.

Appears in An overview of angioedema: Clinical features, diagnosis, and management

ACE inhibitor-induced angioedema



ACE: angiotensin-converting enzyme.

Appears in ACE inhibitor-induced angioedema and in Drug eruptions

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ACEIs / ARBs – ALL RAS INHIBITORS: CONTRAINDICATIONS

- Angioedema related to any cause
- Hypersensitivity / Anaphylaxis
- **Pregnancy:**

Absolutely contraindicated in all trimesters

  - Fetal hypotension
  - Fetal anuria and renal function $\Rightarrow$ Oligohydramnios
  - Fetal lung hypoplasia
  - Skeletal malformations, skull hypoplasia
  - Death of the fetus/neonate

ACEI / ARB Drug-Drug Interactions

- ACEI + ARB
- ACE inhibitor or ARB + direct renin inhibitor

→Increased adverse effects

Potentially life-threatening

hypotension, syncope, hyperkalemia, kidney dysfunction

Concomitant RAS inhibitor therapy is
NOT RECOMMENDED

- K^+-sparing drugs: ENaC inhibitors, MR antagonists, NSAIDs
- NSAIDs also blunt the antihypertensive effects by increasing reabsorption of Na^+ and water.
- Lithium levels may increase requiring lower lithium dose (monitor).

RAS Inhibitors:
Direct Renin Inhibitor (DRI): Aliskiren

$$\text{Angiotensinogen} \xrightarrow{\text{Renin}} \text{Angiotensin I}$$

Aliskiren is a competitive inhibitor of renin. It binds to the active site of the enzyme and prevents access to the substrate, angiotensinogen. Activation of RAAS is blocked.

Aliskiren Properties

Chem	Small non-peptide drug	
PK	• Oral • P-glycoprotein substrate; bioavailability ~3% • Extent of metabolism unknown, may involve CYP3A4 • Excreted mainly unchanged in feces via biliary system • t½ ~24h / 1x daily dosing	CrCl <30 mL/min: Risk of hyperkalemia is increased. Progressive renal impairment may occur. AUC is increased in patients ≥ 65 years.
MOA and effects	Highly potent, selective competitive inhibitor of renin • Dose-dependent decreases in ANG I, ANG II, aldosterone • Decreases BP: Efficacy comparable to other antihypertensive agents • *Adverse effects* as for other RAS inhibitors: hypotension, hyperkalemia, nephrotoxicity, cough, angioedema, pregnancy contraindication	
Use	Hypertension (alternative for HTN treatment if ACEI or ARB contraindicated or not tolerated) • Monotherapy or in combination with other antihypertensive agents but **not** combined with ACE inhibitors or ARBs	

Aliskiren Drug Interactions

Potentially significant interactions may exist, requiring dose or frequency adjustment, additional monitoring, and/or selection of alternative therapy.

- Other antihypertensive agents (beneficial therapeutic use)
- **Strong P-glycoprotein inducers or inhibitors** decrease / increase aliskiren systemic levels
    - ***Itraconazole, cyclosporine, ritonavir should be avoided.***

Combination with ACEI or ARB ***not recommended***:

1. Increased risk of hyperkalemia / hypotension / renal impairment
2. Does not lower the risk of cardiovascular event

Question: You may want to pause the video.

- An 58-year-old non-diabetic person presenting with crushing chest pain is ultimately diagnosed with acute ST-elevation myocardial infarction (STEMI) with anterior infarct. Percutaneous coronary intervention is performed. Post-procedure oral pharmacotherapy initiated prior to discharge consists of low-dose aspirin, clopidogrel, metoprolol succinate, rosuvastatin, and low-dose lisinopril. Blood pressure is carefully monitored.
- The patient is at high risk for what RAS-inhibitor-related adverse effect?

Question: You may want to pause the video.

- The randomized double-blind Aliskiren Trial in Type 2 Diabetes Using Cardiorenal Endpoints (ALTITUDE) assigned 8,561 patients to either aliskiren or placebo as an adjunct to ACE inhibitor or ARB. The primary endpoint was a composite of the time to cardiovascular death or a first occurrence of cardiac arrest with resuscitation; nonfatal myocardial infarction; nonfatal stroke; unplanned hospitalization for heart failure; end-stage renal disease, death attributable to kidney failure, or need for renal replacement therapy with no dialysis or transplantation available or initiated; or doubling of baseline serum creatinine.
- The trial was stopped early after the second interim analysis showing that the addition of aliskiren to standard therapy with RAS blockade in patients with type 2 diabetes at high risk for cardiovascular and renal events is not supported by the data (no benefit in primary endpoints) and might even be harmful.
- Thinking about the mechanism-related adverse effects of RAS inhibition, particularly when RAS inhibitors are combined, what were two toxicities occurring at a significantly higher rate in the aliskiren group compared to the placebo group?

Summary of RAS Inhibitors

Angiotensin converting enzyme inhibitors (ACEI)

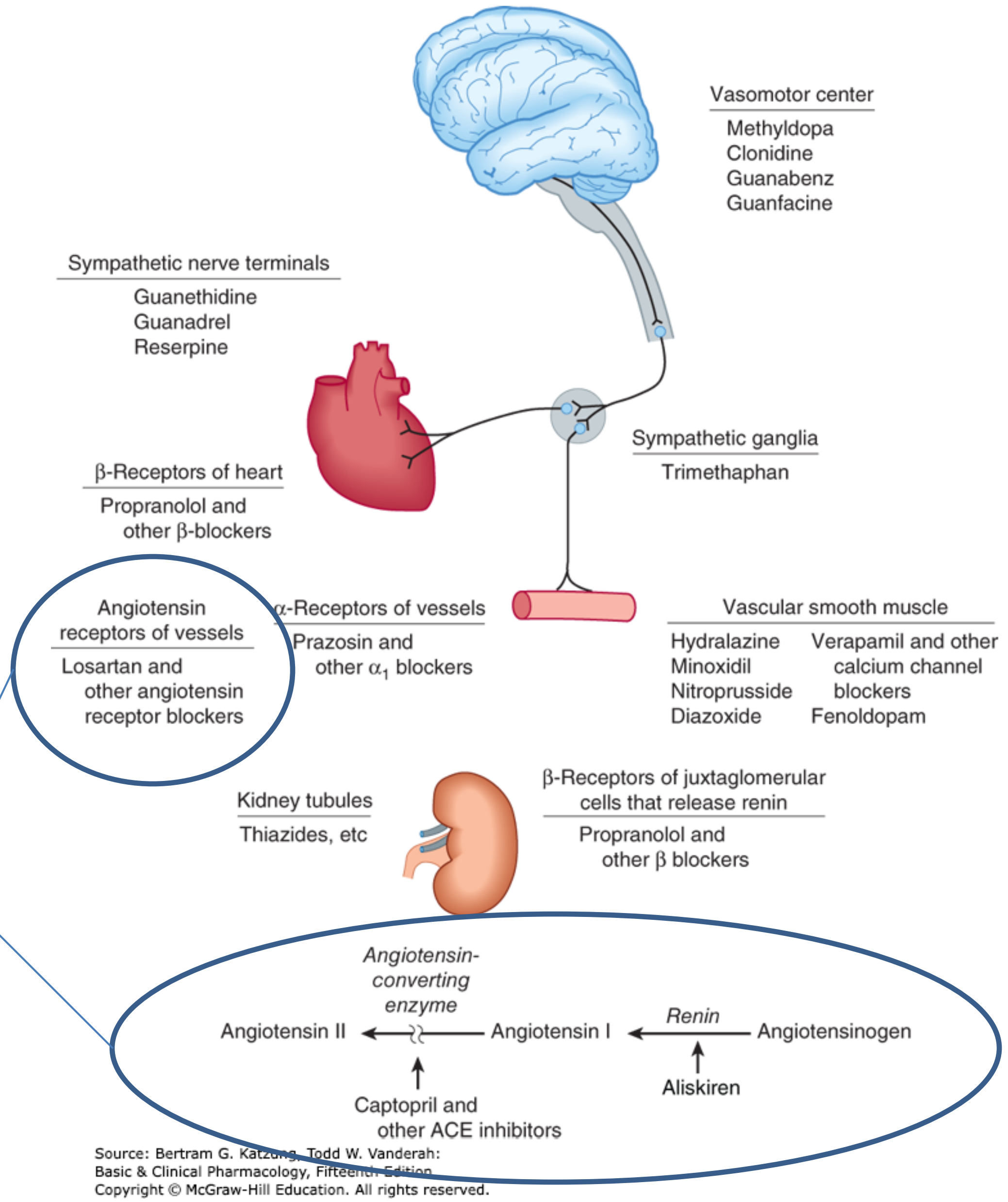
Angiotensin II receptor blockers (ARBs)

Direct renin inhibitor (DRI)

Sites of action of the major classes of antihypertensive drugs.

Most antihypertensive agents work by reducing cardiac output and/or decreasing peripheral resistance.

RAS inhibitors



Citation: Chapter 11 Antihypertensive Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021, Figure 11-3. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595338 Accessed: August 05, 2021

- RAS inhibitors reduce peripheral vascular resistance without reflex sympathetic activation and they decrease aldosterone secretion.
- ACEIs increase vasodilating peptides bradykinin and substance P levels, which may also be associated with causing cough and angioedema. ARBs are associated with a low incidence of cough and angioedema. DRIs can cause these adverse effects.
- ACE inhibitors and ARBs are effective and widely used in the treatment of hypertension, post-acute myocardial infarction therapy, heart failure with reduced ejection fraction, and proteinuric chronic kidney disease. Aliskiren is an option for the management of hypertension. It is not a recommended first-line therapy.
- RAS inhibitors are generally well tolerated.
- RAS inhibitors can cause hypotension; patients with RAAS activation are at high risk. Other adverse effects include modest reduction in GFR, hyperkalemia, and nephrotoxicity. ACEIs are associated with a dry cough and angioedema that may be related to elevated levels of bradykinin and substance P.
- Olmesartan is associated with sprue-like enteropathy with severe diarrhea and weight loss.
- **Combinations of RAS inhibitors should be avoided due to increased risk of hypotension, syncope, renal impairment, and hyperkalemia.**
- RAS inhibitors are contraindicated during pregnancy because of the risk of fetal hypotension, anuria and renal failure leading to oligohydramnios. Fetal malformations and death may occur.

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